Dragos Mihai Sorescu

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SUMMARY

Honours BSc AI student with hands-on experience building and deploying ML systems, from a vision-transformer model achieving ~92% sensitivity to an XGBoost service running on GCP Cloud Run. Currently an AI Research Assistant at Durham College's AI Hub, developing a serverless multi-agent platform. Seeking an AI/ML co-op where I can contribute to production-grade systems.

EDUCATION

Durham College Oshawa, ON

Honours Bachelor of Science in Artificial Intelligence *Expected April 2027*

- GPA: 3.81 | Winter 2025 Honour Roll
- Relevant Courses: Machine Learning, Computer Vision, Natural Language Processing, Statistics, Linear Algebra, Data Structures and Algorithms, Database Management Systems, Systems Analysis and Design, Computer Architecture

SKILLS

Languages: Python, SQL, TypeScript

ML / AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, LangChain, LangGraph, Anthropic SDK

Data: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Web: FastAPI, SvelteKit, Streamlit

Cloud & Infra: AWS (EC2, S3, Lambda, Bedrock, API Gateway), GCP, Docker, Linux

Databases: PostgreSQL, MySQL, Supabase

Tools: Git, Pytest, Locust, Ragas, Tableau, Power BI

PROFESSIONAL EXPERIENCE

AI Research Assistant May 2026 – Present

Durham College AI Hub *Oshawa, ON*

- Built the agentic core for a multi-persona LLM chatbot on AWS Lambda and Bedrock, supporting persona routing, multi-turn context, and SSM-managed prompts; load-tested to 1.8s p95 latency with Locust.
- Shipped a safety redline layer that detects high-risk language before LLM classification and triggers immediate human escalation, with deterministic matching and a 224-line unit test suite.
- Implemented confidence-based routing with tiered thresholds, safety-intent bypasses, and multi-turn escalation detection across conversation history.
- Authored the handoff and escalation trigger spec for six intent branches, defining automatic vs. suggested routing and criteria for safety validation.

PROJECTS

Financial News Agent | Python, FastAPI, Claude API, Finnhub, yfinance, Supabase, SvelteKit | Repository

- Built an agentic financial assistant with Claude API and autonomous tool use, allowing the model to call price, news, financial, and insider-trade tools based on user intent.
- Secured FastAPI endpoints with Supabase JWT authentication, row-level security, per-IP rate limiting, and input sanitization.
- Built a SvelteKit frontend with SSE streaming, real-time tool-call progress, and modular market and portfolio views.

Medical Image Analysis with Vision Transformers | Python, PyTorch, Scikit-learn, Pandas, OpenCV, Matplotlib, Seaborn | Repository

- Benchmarked 8+ vision architectures, including DINOv2, ViT, SwinV2, ResNet, and YOLO, for binary skin cancer classification using stratified 5-fold cross-validation.
- Achieved ~92% sensitivity at a clinically relevant threshold using ablation studies, freezing strategies, and ensemble evaluation.
- Built preprocessing, augmentation, and training pipelines for a quality-controlled 906-case skin-lesion dataset.

Temperature Trend Prediction App | Python, Pandas, Scikit-learn, XGBoost, FastAPI, Streamlit, Docker, GCP | Repository

- Trained an XGBoost regression model on historical weather data, using regularization and cross-validation to reduce overfitting.
- Deployed the model as a containerized FastAPI service on Google Cloud Run with a Streamlit interface for public access.